

CropYield Predictor

Agricultural Intelligence & Analytics Platform

Final Analysis Report

End-to-end data validation, exploratory data analysis, domain feature engineering, cross-validated model benchmarking, hyperparameter tuning, model diagnostics, and Streamlit dashboard delivery.

25,932 validated records | 101 areas | 10 crop categories | 1990-2013

This report documents the completed project artifacts and the measured outputs included in the submission package.

1. Executive Summary

The project predicts crop yield in hg/ha from Area, Item, Year, rainfall, pesticide use, and average temperature. The deployable model is a Tuned Random Forest Regressor inside a pipeline containing training-set IQR capping, agricultural feature engineering, a ColumnTransformer, and the regressor. The held-out test set contains 5,187 records.

Held-out performance: $R^2 = 0.9813$, $MAE = 4,589.79 \text{ hg/ha}$, $RMSE = 11,654.74 \text{ hg/ha}$. The R^2 result is above the project brief target of 0.80.

The analysis separates descriptive pooled relationships from causal interpretation. Crop identity and geographic indicators account for substantial group-level differences, while raw macro-climate correlations with the pooled target are modest.

Included deliverables: executed 21-visualization EDA notebook, model-comparison artifact, RandomizedSearchCV tuning evidence, joblib model, feature importance, residual diagnostics, Streamlit dashboard source, local experiment artifacts, final PDF, and a five-minute caption-led video walkthrough.

Dataset Profile

Metric	Value
Records	25,932
Columns	7
Areas	101
Crops	10
Years	1990-2013
Missing values	0
Duplicates after cleaning	0

2. Data Quality and Reproducible Pipeline

The raw agricultural data were validated for schema, missing values, numeric bounds, crop categories, duplicates, and potential post-harvest leakage columns. The processed export contains 25,932 rows and 7 columns with zero missing values and zero duplicates.

Outlier treatment is implemented as an IQR capper inside the scikit-learn pipeline. The cap bounds are fit only on training data; during cross-validation the transformer is cloned and fit separately in each fold. The target column is not capped.

Pipeline order: IQR Capper -> AgriculturalFeatureEngineer -> ColumnTransformer -> Regressor. Numeric inputs use median imputation and standard scaling. Area, Item and rainfall_category use most-frequent imputation and one-hot encoding with unknown-category handling.

The saved model file is a full sklearn Pipeline and was loaded without retraining to generate a sample prediction for India / Rice, paddy.

Outlier Audit

Feature	Potential IQR outliers	Treatment
Year	0 (0.00%)	Training-set IQR capping
average_rain_fall_mm_per_year	0 (0.00%)	Training-set IQR capping
pesticides_tonnes	1,262 (4.87%)	Training-set IQR capping
avg_temp	16 (0.06%)	Training-set IQR capping

3. Feature Engineering and Modeling Design

Four domain-derived features are engineered inside the model pipeline: $\text{hydrothermal_index} = \text{rainfall} / (\text{temperature} + 10)$, rainfall_category , $\text{pesticide_log} = \log_{10}(\text{pesticides})$, and $\text{temp_rainfall_interaction} = \text{rainfall} * \text{temperature} / 1000$.

Five distinct model architectures are benchmarked using the same shuffled 5-fold KFold design: Linear Regression, Ridge Regression, Decision Tree, Random Forest, and Gradient Boosting.

The final model is a Random Forest using $n_estimators = 50$, $max_depth = 25$, $min_samples_split = 2$, and $min_samples_leaf = 1$.

The project also records a RandomizedSearchCV experiment for Random Forest. The documented mean CV R2 moves from 0.9827 before tuning to 0.9830 after tuning, a small 0.03% relative improvement.

Interpretation note

The stronger tree-model results indicate that the dataset contains non-linear and categorical structure that linear baselines do not capture as well. These metrics describe predictive performance and should not be interpreted as evidence of causal agricultural effects.

4. Model Comparison, Tuning and Error Diagnostics

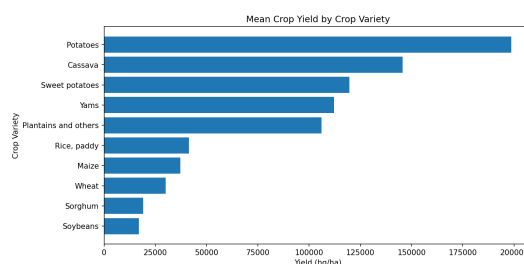
Model	CV R2	CV MAE	CV RMSE	Test R2
Linear Regression	0.7481	29,617	42,409	0.7484
Ridge Regression	0.7461	29,339	42,583	0.7458
Decision Tree	0.9690	5,432	14,864	0.9700
Random Forest	0.9827	4,947	11,089	0.9810
Gradient Boosting	0.9397	13,582	20,752	0.9349

Final error summary: normalized MAE = 5.85% of the mean actual yield. 75.03% of held-out rows are within +/-10% relative error and 87.83% are within +/-20% relative error.

Feature importance is reported for the final Random Forest. The largest individual encoded feature is Item_Potatoes; other crop, area and numeric environmental variables also contribute. Feature importance is a model explanation signal, not a causal ranking.

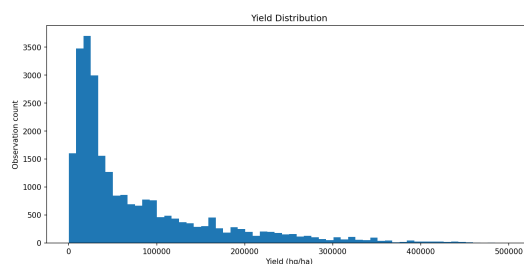
5. EDA Visualizations 1-3

Figure 1: Mean yield by crop variety



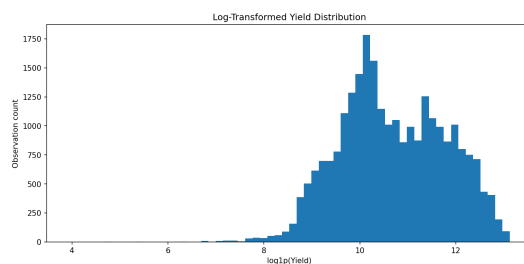
Potatoes has the highest mean yield at 198,666 hg/ha among the 10 crop categories, while Soybeans has the lowest mean.

Figure 2: Yield distribution



The pooled target is strongly right-skewed, motivating robust tree-based modeling and careful treatment of extreme feature values.

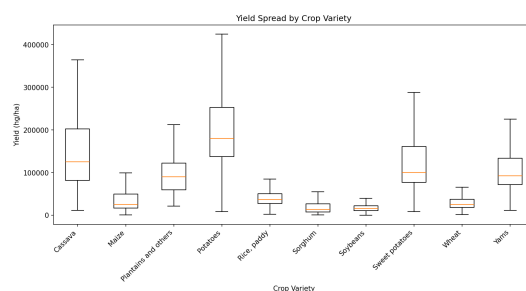
Figure 3: Log-transformed yield distribution



The log transform compresses the long right tail and makes the central distribution easier to inspect.

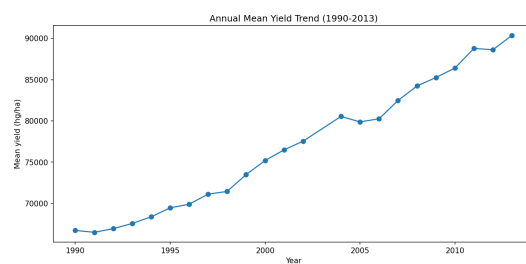
5. EDA Visualizations 4-6

Figure 4: Yield spread by crop



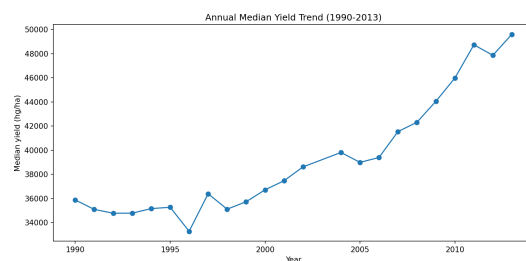
Yield ranges differ materially by crop, indicating that crop identity is a dominant explanatory variable.

Figure 5: Annual mean yield trend



Mean pooled yield rises from about 66.7k hg/ha in 1990 to about 90.4k in 2013, with small year-to-year dips in a few years.

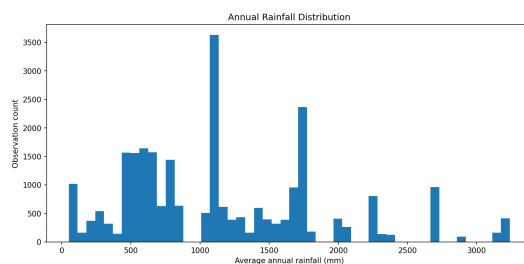
Figure 6: Annual median yield trend



The median series follows the broad upward pattern while reducing the influence of high-yield crop categories.

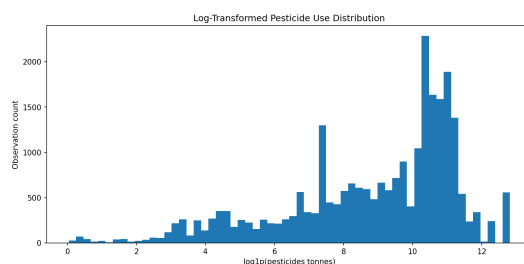
5. EDA Visualizations 7-9

Figure 7: Annual rainfall distribution



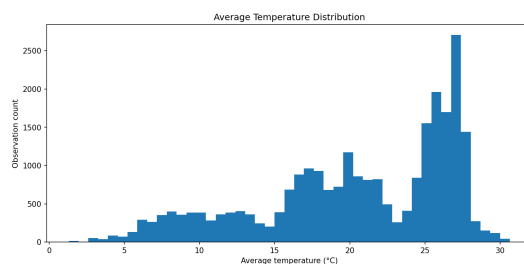
Rainfall spans 51 to 3,240 mm/year, covering multiple climatic regimes used later for categorical feature engineering.

Figure 8: Log-transformed pesticide distribution



Pesticide input is highly skewed; log1p provides a numerically stable representation for EDA and modeling.

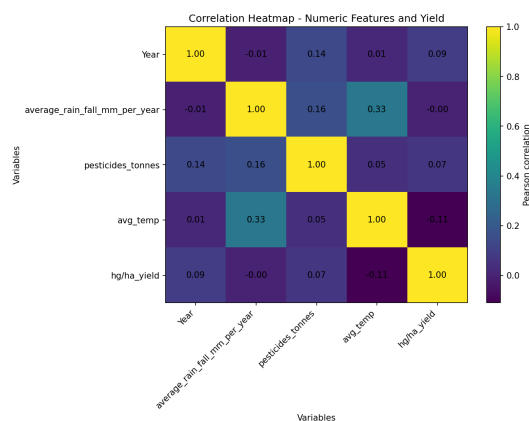
Figure 9: Average temperature distribution



Average temperature ranges from 1.3°C to 30.65°C, indicating wide geographic coverage.

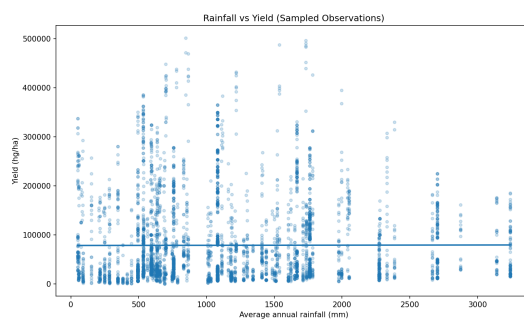
5. EDA Visualizations 10-12

Figure 10: Numeric correlation heatmap



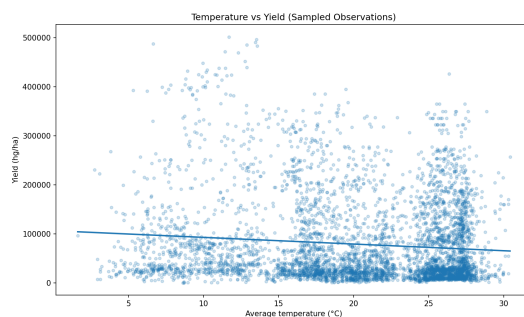
Raw linear correlations with yield are modest for rainfall, pesticides and temperature, because the pooled target varies strongly by crop and country.

Figure 11: Rainfall vs Yield (Sampled Observations)



The pooled relationship is weak because crop type and geography create substantial between-group variation.

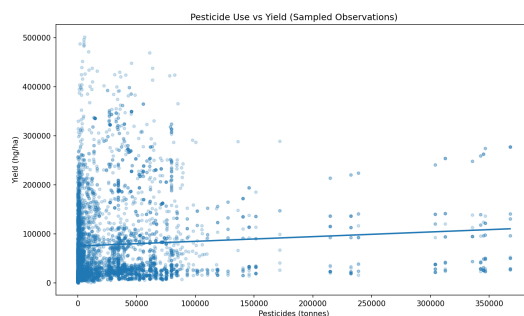
Figure 12: Temperature vs Yield (Sampled Observations)



The raw pooled association is slightly negative, but the scatter shows strong clustering across crop categories.

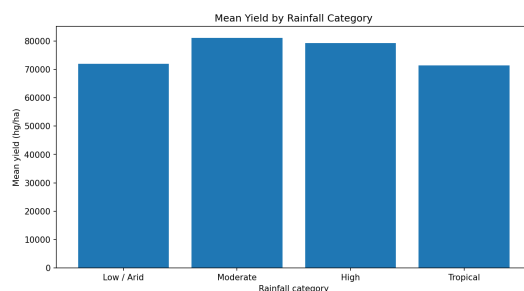
5. EDA Visualizations 13-15

Figure 13: Pesticide Use vs Yield (Sampled Observations)



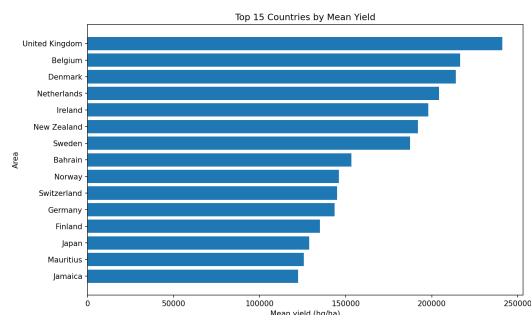
Pooled observations show a modest positive raw association, with large dispersion driven by geography and crop mix.

Figure 14: Mean yield by rainfall category



Average yield differs across rainfall regimes; the categorical feature provides a simple way to represent non-linear climate bands.

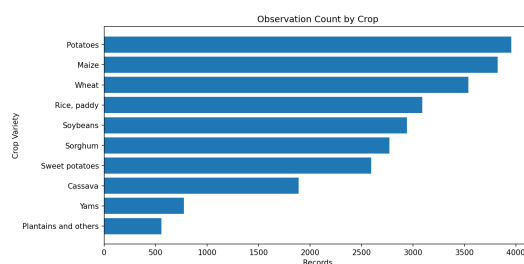
Figure 15: Top countries by mean yield



The highest pooled country means reflect the mix of crop types and production systems represented in each area, so rankings should not be interpreted as causal country effects.

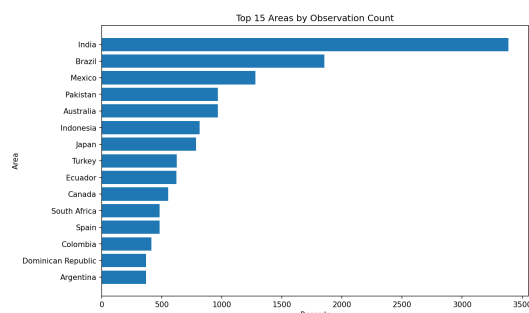
5. EDA Visualizations 16-18

Figure 16: Observation count by crop



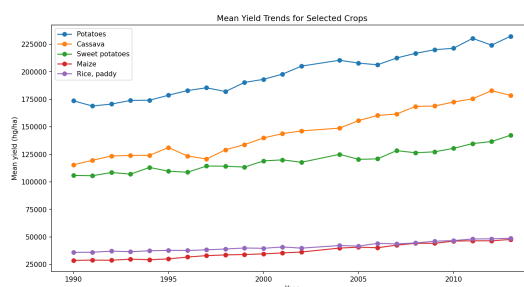
The dataset is unbalanced across crop categories; potatoes and maize contribute more observations than several other crops.

Figure 17: Observation count by area



India contributes the largest number of observations among the areas in this dataset, so pooled analyses should account for unequal group sizes.

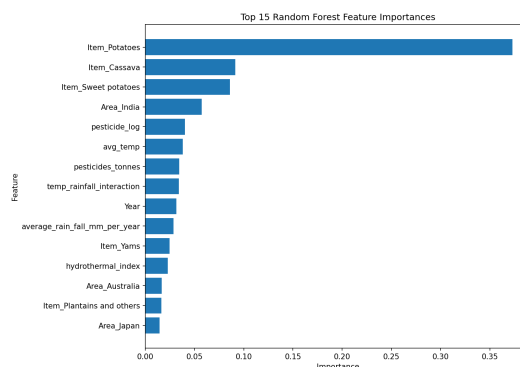
Figure 18: Selected crop yield trends



Different crops exhibit distinct baseline yield levels and trajectories; this motivates retaining crop identity in the predictive pipeline.

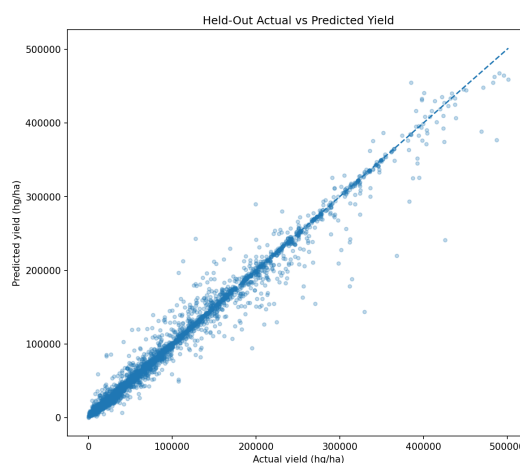
5. EDA Visualizations 19-21

Figure 19: Top feature importances



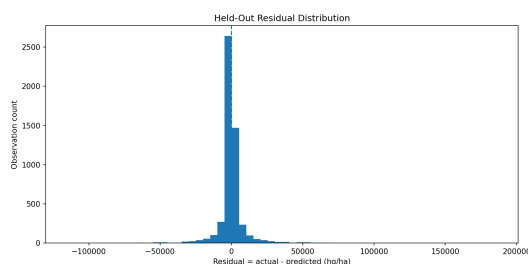
The final Random Forest places high importance on crop indicators such as Item_Potatoes, followed by other crop, area and numeric features.

Figure 20: Held-out actual vs predicted yield



Predictions cluster around the 45-degree reference line, consistent with the held-out R^2 of 0.9813.

Figure 21: Residual distribution



Residuals are centered near zero; the remaining spread reflects observations that are harder to predict from the available macro-level features.

6. Delivery and Reproducibility

The Streamlit dashboard contains Data & EDA, Prediction, Model Comparison, and Feature Importance views. The prediction form sends raw input fields directly to the serialized pipeline without retraining.

The repository includes executed notebooks, model metadata, local experiment artifacts, requirements, tests, the final PDF report, and a five-minute caption-led video walkthrough. Install dependencies from requirements.txt before running the dashboard or notebook stack.

Conclusion: the project package now contains the requested end-to-end workflow, with the main remaining deployment requirement being installation of the declared runtime dependencies in the target environment.